

Comments & Replies: Discussion Forum Interaction Networks Stimulated by Peer-Review Assignments

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Abstract—Social aspects of in-person learning activities such as peer review, group work, and class discussion can better develop connections both between learner and course content and between learner and learner. Online courses however, struggle to maintain these experiences of in-person learning, especially at larger scales in massive open online courses (MOOCs). The lack of a learner network of support and or of collaboration can reduce learner engagement and learner outcomes in an online course. In this work we provide an analysis of the types of connections learners create within MOOCs via interactions in the collaborative discussion forum platform, CrowdLearn (CL). The CL platform is designed to enable peer-review of open-ended assessments that are commonly used to provide a formative learning experience in upper-level courses. Across two instances of an undergraduate-level manufacturing processes online course, a total of 67 learners interacted over 1600 times with the CL platform. We analyzed the peer review and discussion interactions within four learner CL networks, in which, we identified three distinct learner types. Learner types and network structures were used in conjunction with learner outcome data to highlight various trends regarding the effects of using such a platform in MOOCs. We identify relationships that highlight challenges of implementing collaborative discussion forum platforms at scale. Further, we suggest potential areas of future work to overcome some of these challenges.

Keywords—MOOCs, edX, CrowdLearn, discussion forums, networks, learner outcomes, collaborative learning

I. INTRODUCTION

Massive open online courses (MOOCs) provide an accessible and individual learning environment to tens of thousands of learners worldwide [1]. With hundreds of learners accessing course content at different times, many of the formative and collaborative learning experiences unique to in-person classrooms are not transferable to these online formats – leading to a shift in how course contents are designed and disseminated [2-4]. A sizable benefit to participating in MOOCs is that learners can take in information on their own and in their own ways while maintaining fit to their schedules. This individually-motivated form of learning can be empowering; however, it can also be isolating [5-8]. Outside of the online classroom, discussion forums have enabled the creation of communities via sharing content and discourse [9]. This sense of community has been shown to greatly increase a person's engagement with content [10]. As such, discussion forums have been integrated into MOOCs to help de-isolate the online learning experience [11].

In one common MOOC platform, edX (edX LLC., Maryland, USA), discussion forums are available with each lecture and assignment page to facilitate learners to ask questions and to share ideas about the content they are

working with. The presence of a discussion forum alone does not fully address the limitations of MOOCs [12]. It is important to design collaborative learner-to-learner experiences into the course structure itself. Many higher-level in-person courses incorporate group projects where learners work together on a solution and present their collective outcomes to other groups within the class. Experiences like these have greatly improved learner outcomes with course content and helped learners create scholastic social networks [13-17].

CrowdLearn (CL) is a tool hosted by FeedbackFruits (Amsterdam, Netherlands) to enable open-ended and formative assessment in MOOCs. CL achieves this by involving the learners themselves as graders on their assignments [18]. Within the CL platform, learners can submit assignments as PDFs and then other learners are able to interact with the submissions by leaving comments, replying to comments, and or upvoting submissions, comments, and replies alike. The submissions themselves can be viewed with all comment history attached, like that of a word processing interface. The rest of the CL interface is designed like a discussion forum, where each learner's submission is a post and all communications related to the post are shown in a dropdown interface below the post.

This work focuses on the networks developed within CL assignments of the edX course – Fundamentals of Manufacturing Processes. Network and learner connectedness are analyzed in relationship with learner outcomes. We identify relationships that highlight challenges of implementing discussion forum platforms at scale. Further, we suggest potential areas of future work to overcome some of these challenges.

II. METHODS

A. Demographics

67 learners participated in the four CL assignments. 28 countries were represented. Of the learners that provided the following information: 87% were male, 13% were female; as their highest education – 5% had a high school diploma, 2% had an associate's degree, 55% had a bachelor's degree, 33% had a master's degree, and 5% had a doctorate.

B. Structure of Fundamentals of Manufacturing Processes Course

Fundamentals of Manufacturing Processes is the online version of a like-named undergraduate residential course. The online course is hosted on the edX platform. The aim of this course is to highlight the diversity, complexity, and importance of manufacturing processes in our daily lives. This edX introduces learners to the concepts of advanced

manufacturing, with emphasis on the following: manufacturing processes, equipment and automation, variation and quality, creating value at scale, and design for manufacturing.

The fall 2021 (F21) and spring 2022 (S22) runs of 2.008x included two peer-review discussion forum “challenges” via the CL platform. Only learners that paid for the certificate of the edX course (verified learners) were able to access these challenges. The first challenge was focused on the concepts of casting and will be referred to as Assignment I throughout the rest of this work. Assignment I tasked learners to examine two cast objects and to answer questions related to the objects. The second challenge was focused on the concepts of cost and will be referred to as Assignment II throughout the rest of this work. Assignment II tasked learners to examine a manufactured tablet device and to answer questions pertaining to this device. Both assignments were each worth 10% of the learners’ final grades for the course and were presented with the following instructions:

Answer the questions – study the casting objects/tablet components, answer the questions, and post your responses as a PDF on the open discussion tool.

Review/comment on a few of your peers – you will be assigned to review the submissions of a few peers. After reading their submissions, you can post comments on their work.

Open discussion – you will join the whole community of verified learners to discuss each other’s submissions.

Peer citation – Select at least one comment from your fellow learners that impacted your learning.

Reflect – you will write a quick reflection on what you learned after discussing with your peers.

Each learner had to submit a piece of work in response to the assignment to gain access to the CrowdLearn experience. At random, each learner was assigned to up to four other learners and was required to grade and comment on those learners’ work. Learners were given access to the answer key to the assignment after submission. Learners were encouraged to leave at least two comments on their assigned peers’ work. After this prescribed interaction, all learners’ submissions with their peer-reviewers’ comments were opened to the rest of the CrowdLearn users in an open discussion forum format. The open discussion was where learners could comment/reply to comments on other learner’s submissions that they were not assigned to review. These interactions were not required but encouraged. After open discussion, learners were asked to reflect on how another learner’s comment on their submission and on how the overall open discussion experience affected their learning of course content.

The data used to generate all the network analysis within this work was collected from the learner interactions within the peer-review and open discussion steps of each assignment instance. Of note, this work contains analysis of only the comment and reply interactions within the CL networks – no upvote or text analysis is within this work.

C. Collating CrowdLearn Interactions

User interaction data was exported from the CrowdLearn instructor interface for each instance of the course challenges. The exported CSV files included all documented interactions

sent by users and included the peer review assignments amongst users. In a separate XLSX file, user interaction data was collated and anonymized. Users were anonymized by replacing all calls to user defined usernames with their respective random 32-character submission file ID. The anonymized data from each of the challenge instances was organized into a master XLSX file, which was contained in separate sheets per challenge instance. The data within each sheet contained a tab delineated line of all information pertaining to each interaction recorded in the CrowdLearn instance. Each line contained the relevant submission user ID, commenting/replying/upvoting/ user ID and the respective receiving user ID, and all assigned user peer IDs.

D. Network Generation

Python 3.10.5 was used to generate all analysis and network representation within this work. The master data file was iterated through to create python dictionaries for each column of information within each sheet of the XLSX file. The dictionaries were then called throughout the code to provide key information to input into the vis.js ‘Network’. Network is a visualization package that consists of nodes and edges, in which the nodes can be clustered depending on simulated physics parameters like: gravity, mass, and spring constants [19, 20]. This package was chosen for its visualization flexibility and support for custom node shapes, styles, colors, and more.

A network was generated for each instance of CrowdLearn (four). A node was defined for each unique user submission, this was also how user IDs were defined. Node size and mass were increased linearly based on how many interactions the respective node sent into the network. Connecting edges were defined by each unique directional interaction instance. Repeat interactions of the same two nodes and to-from directions were not visualized to avoid congested network visualizations. Repeat interactions of this type were represented by the length of the connecting edge between the two nodes – the more interactions a node sent to another node, the shorter the length of the connecting edge. Edges were set discretely and with high spring constants to maintain rigid directional connections between nodes. Gravity and repulsion were dependent on overall interactions sent from each node. For each interaction sent, the node’s mass and size were increased by 1 increment. Node masses started from a value of zero, while node sizes were set to a range between the total number of nodes in the network and the total number of nodes in the network times two. Nodes that sent the least number of interactions were given the highest repulsion. Nodes that sent more interactions gained gravity (less repulsion). Edges between assigned peers were filtered to not be included unless the node pairs had more than the requested two interactions between them. Only the user interactions of submitting users were considered and represented within a network.

E. Types of Interactions

There were four interaction types recorded via the CrowdLearn interface – Submission, Commenting, Replying, Upvoting. Submission was required to take part in the challenge and peer review discussion forums. Upon submitting the required materials, users can comment, reply, or upvote within the discussion forums. Commenting was required at least twice between each assigned peer for the review process in the assignment and was represented by black connecting edges. Replying was not a required

interaction and was represented by red connecting edges. Upvoting was the most passive interaction not required and would be represented by green connecting edges. Since commenting at least two times was suggested between assigned peers, only interactions greater than two were visualized/counted for assigned peer interactions.

F. Network Metrics & Statistical Analysis

Two metrics were calculated from the generated network visualizations and are defined below:

$$\text{Avg. Connections per Learner} \equiv \frac{C_N}{N} \quad (1)$$

In which, C_N is the total number of unique learner-to-learner connections present in a given network, N is the total number of learners in a given network. This metric was utilized to measure how connected a network was when comparing the four CL instances.

$$\text{Connectedness Level} \equiv C_L = \left\lceil \frac{C_n}{C_{min}} \right\rceil \quad (2)$$

Above represents the number of unique connections a given learner had to their network (C_n) divided by the minimum C_n recorded amongst all CL instances (C_{min}), rounded to the nearest integer. This metric was used to bin learners into comparable levels from one to five, with level one being the lowest level of connectedness and level five being the highest (this level includes all $C_L > 4$). Bailey et. al. provided the basis and inspiration for a type of connectedness index [21]. Their work focused on the overall connections people can make through a collection of social networks within the social media platform, Facebook.

The following network statistics and superlatives were defined directly from the network datasets: most frequently sent interaction type, most connected learner, most popular learner, most engaged learner, biggest commenter, biggest replier, and biggest upvoter. The most popular learner was defined by the user that received the most interactions. The most engaged learner was defined by the user that sent the most interactions. The directly acquired superlatives were used to connect learner behaviors to challenge specific and overall course outcomes.

Single factor analysis of variance (ANOVA) tests were used to validate the statistical significance between learner outcome data and learner connectedness levels (C_L). One-tailed heteroscedastic t -tests were used to determine the strength of using learners' connectedness levels (C_L) as a grouping method for learner outcome data. For both statistical analysis methods, P-values were reported. P-values less than 0.05 were considered a confirmation of hypothesis.

III. RESULTS & DISCUSSION

A. Types of Learners

Three learner types were classified – Inactive, Reactive, and Active (**Figure 1**). Inactive learners were those that submitted the required materials and participated in the minimum required interaction during the required peer review discussions. These inactive learners only interacted with assigned learners in the peer review up to the requested two times. Reactive learners were those that have submitted the required materials and participated beyond the required two interactions; however, most interactions past the requested two were in reply to other learner interactions. Active learners were those that submitted the required materials and

participated beyond the required two interactions; however, these learners initiated most of their interactions via new comments. In the related work by Hao et. al., they defined learner classifications: “boundary spanners”, “content contributors”, and “opinion leaders”; which respectively share similarities with our Inactive, Reactive, and Active classifications [22]. Further in prior work by Steinmaurer et. al., learner classifications such as “core users”, “help givers”, “help seekers”, and “isolated user” were defined [23]. We chose to diverge from these classifications since they required a level of sentiment analysis not present in this work. Figure 2 shows that learners in all CL instances were predominantly Active. The assignment directions to leave at least two comments on assigned peer learners' content may have had an effect here.

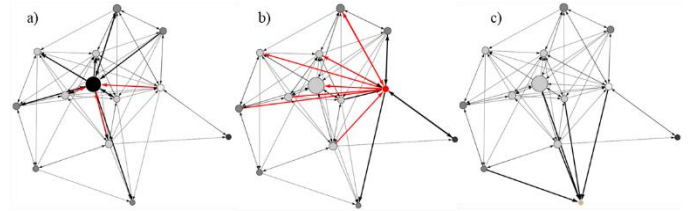


Figure 1: a) An Active learner that comments on many others' content, b) A Reactive learner that mostly replies to interactions with them, c) An inactive learner – never sent any interactions. (S22 Assignment II shown)

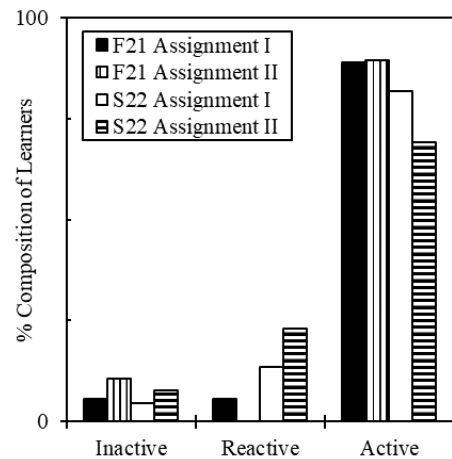


Figure 2: Largely, learners were active within each network.

B. Learner Network Characteristics

Average connections per learner (1) was defined as the total number of established unique learner-to-learner connections (C_N) divided by the total number of learners within the network (N). There was a clear positive correlation between the number of learners and average connections per learner (**Figure 3**). Networks that had a larger N resulted in a higher average connections per learner. This positive relationship followed a logarithmically increasing fit, which suggests that an asymptotic maximum of connections per learner may exist as the number of learners increases.

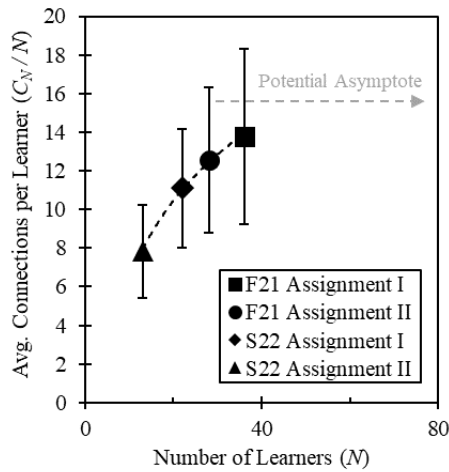


Figure 3: The relative size of each network showed a strong correlation to overall network connectedness.

Irrespective of network size, the ratio of comments sent to replies sent loosely ($R^2 = 0.73$) followed a parabolic fit (**Figure 4**) – suggesting that there may be a local maximum where the ‘right’ number of comments to replies leads to the highest average connections per learner. If this parabolic relationship proves to be true as more CL instances are analyzed, the following two concepts would become apparent. First, discussion forums in which users mostly reply to comments quickly get siloed and users end up mostly interacting with a group of the same users. This phenomenon has been explored in prior work provided by Thomas et. al. and Chen et. al. [24, 25]. Second, discussion forums in which users mostly comment may disincentivize users from continuing to reach-out since their efforts are rarely getting the supportive interactions of replies.

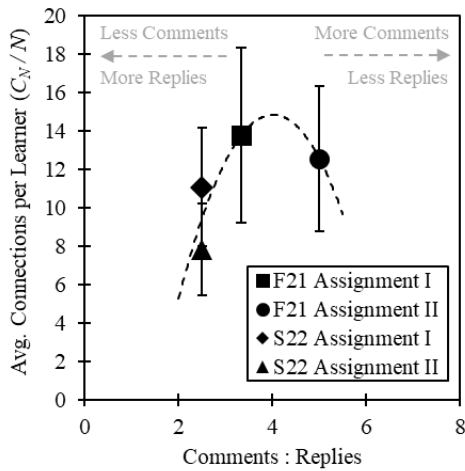


Figure 4: A local maximum of comments to replies demonstrates a potential ‘best’ ratio to maximize connections per learner.

Connectedness level (2) represents a learner’s number of connections divided by the minimum number of connections a learner had amongst all CL instances, rounded to the nearest integer. This sets connectedness level 1 as the minimum number of connections and level 2 as the range of two times the minimum number of connections, and so on. Each network was partitioned into five connectedness levels. Generally, learners that fell into the fourth and fifth

connectedness levels were found towards the centers of the networks and represented the learners with the higher connectedness levels (**Figure 5**). $30 \pm 19\%$ of learners were in the connectedness level 5. $54 \pm 13\%$ of learners populated the next two connectedness levels (4 and 3). $17 \pm 9\%$ of learners were in connectedness levels 2 and 1. Within each course instance from the first CL assignment to the next (**Figure 5a-b** and **Figure 5c-d**), learners decreased their connectedness levels. Many higher connectedness level learners migrated down a level when engaging in the second CL assignment. This decreasing of connectedness level suggests that after heavily engaging and doing well on the first CL assignment, learners may be likely to adjust their amount of effort with the future CL assignment.

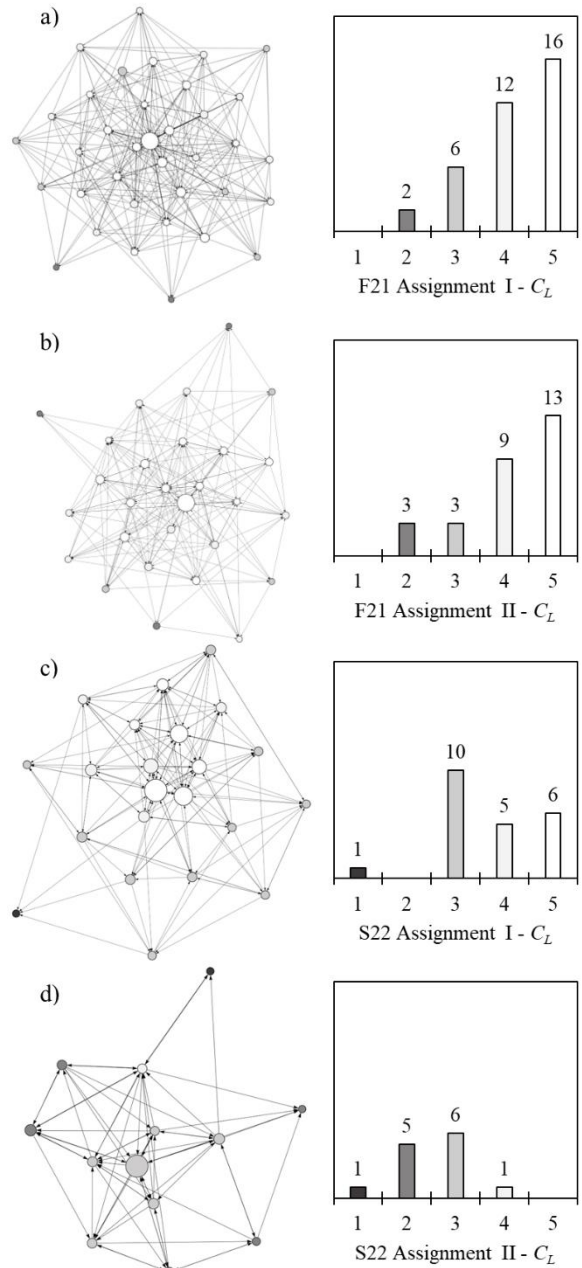


Figure 5: All generated networks from each CrowdLearn instance and their respective C_L distributions.

C. Connections to Learner Outcomes

In Figure 6 below, the course and assignment outcomes are averaged per each connectedness level across both the F21 and S22 runs of the course. A positive trend was observed between the grades of learners and their respective connectedness levels. The strongest correlation was between learner connectedness levels and assignment grades, with a P-value determined via ANOVA of $2.6 \cdot 10^{-7}$. This suggests that learners that engaged more in the assignment discussions would garner a better assignment grade. Course grades correlated with an ANOVA P-value of 0.031. The positive trend between connectedness levels and course outcomes could suggest that learners that engage more in the CL discussion forums are more likely to perform better by the end of the course – highlighting an importance to aim to increase learner connectedness with user interface or assignment design changes. These statistically significant relationships were aligned with the moderate relationships reviewed by Doo et. al. [26].

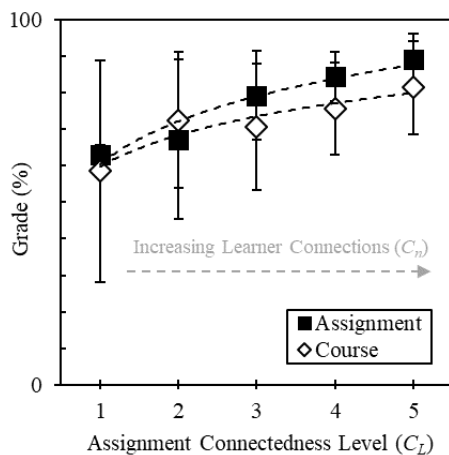


Figure 6: More connected learners end up performing better in both assignment outcomes as well as course outcomes.

Comparing connectedness levels of assignment grades, the only t -test results with P-values less than 0.05 were levels 4 and 5. The t -test for level 3 resulted in a P-value of 0.066 – highlighting an onset of lower levels beginning to overlap in mean grades. The t -test of the CL grades of level 2 compared to the CL grades of level 1 resulted in a P-value of 0.29 showing that level 2 and 1 were very close to each other. All t -test results for connectedness levels of course grades except for level 5 were above a P-value of 0.05, demonstrating the looser connection of assignment connectedness levels to course grades described above.

IV. CONCLUSIONS & FUTURE WORK

In this work, we provided a network analysis of four CrowdLearn (CL) hosted peer-review discussion forum assignments within the undergraduate level massive open online course (MOOC) – Fundamentals of Manufacturing Processes. Network and learner connectedness were analyzed in relationship with learner outcomes. A relationship with course size highlighted a potential metric to classify the success of a tool aimed at creating connected class experiences at scale in MOOCs. This suggests that depending on how ‘good’ the implementation of the discussion forum

platform is in a MOOC, the overall connectedness at scale will either increase or decrease. Improving network connectedness was shown to positively affect learner outcomes. Continued analysis of learner interaction data remains. Analyzing upvoted interaction data could lead to the description of a separate learner class that primarily uses the more passive interaction. Sentiment analysis of text from each comment and reply can be used to tell which types of messages lead to more responses and more network connections. Filtering interaction quality based on comment and reply lengths may help understand the strength of various connections learners make within a network. Overall, learner connectedness promoted improved outcomes and can be used as a compelling metric to design more social experiences in MOOCs.

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